

Project Design Phase
Problem – Solution Fit Template

Date	28 June 2026
Team ID	Not Given
Project Name	Rising Waters
Maximum Marks	2 Marks

Team Members	Skill Wallet Team ID's	Roll Numbers (College: ACET)
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Problem – Solution Fit Template:

The **Rising Waters** project addresses the challenge of inaccurate and delayed flood forecasting by using machine learning to analyze historical weather data. It provides timely and reliable flood risk predictions through a Flask web application, enabling authorities to make informed decisions, improve disaster preparedness, and reduce the impact of floods on communities.

Purpose:

- ☐ **Provide accurate and timely flood predictions** using machine learning to support early warning systems and reduce the impact of floods.
- ☐ **Enable disaster management authorities and meteorologists** to make faster, data-driven decisions for evacuation planning and emergency response.
- ☐ **Improve disaster preparedness** by analyzing historical weather data and identifying potential flood risks before they occur.
- ☐ **Deliver an easy-to-use web application** that allows users to input weather parameters and instantly receive flood risk predictions.
- ☐ **Enhance public safety and resource allocation** by helping authorities prioritize high-risk areas and deploy emergency resources efficiently.
- ☐ **Support scalable cloud deployment** through IBM Cloud, making the flood prediction system accessible, reliable, and available from any location.

Template:

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) Disaster Management Authorities Municipal Corporations Flood-Prone Communities Emergency Response Teams NGOs and Local Governments	6. CUSTOMER CONSTRAINTS Limited real-time data. Budget constraints. Poor internet connectivity. Lack of technical expertise. Delayed communication.	5. AVAILABLE SOLUTIONS Weather forecast apps Manual monitoring Government alerts News channels SMS warning systems	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS Predict flood risk before heavy rainfall. Receive timely alerts. Protect lives and infrastructure. Plan evacuations efficiently. Reduce economic losses caused by floods.	9. PROBLEM ROOT CAUSE Inaccurate local flood prediction. Delayed warning systems. Scattered weather data. Lack of AI-based analysis. Poor coordination between agencies.	7. BEHAVIOUR Monitor weather forecasts. Check news frequently. Contact local authorities. Share updates through social media. Prepare emergency kits only after warnings.	
Identify strong TR & EM	3. TRIGGERS Heavy rainfall forecasts. Rising river water levels. Weather department warnings. Monsoon season. Historical flood patterns.	10. YOUR SOLUTION Rising Waters is an AI-powered flood prediction and early warning platform that combines weather forecasts, rainfall data, river levels, and historical flood information to predict flood risks, generate timely alerts, and help authorities and citizens take preventive action before disasters occur.	8. CHANNELS of BEHAVIOUR 8.1 ONLINE Mobile App SMS Alerts WhatsApp Government Websites Social Media	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER Before: Fear, uncertainty, panic, lack of preparedness. After: Confidence, preparedness, safety, peace of mind.		8.2 OFFLINE Local Announcements Community Volunteers Emergency Control Rooms Police & Municipality	

References:

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>